

# Heisenberg Uncertainty Relation Derived as a Special Limiting Case of a Single Elementary Phase Cycle (v.02).

Creator: **Hennadii Kovalov** .

Contact: [g.kovalov@gmail.com](mailto:g.kovalov@gmail.com)

## Abstract

The proposed geometric model is not a return to 19th-century classical mechanics, but rather builds on the latest trends in Planck-scale deterministic cellular automata ('t Hooft, 2021), the theory of internal quantum clocks (Donadi, 2023), and the nonlinear quantum dynamics of asymmetric rotors (Micheli and Stone, 2022). The model directly resolves the topological crisis of describing light nuclei as saturated manifolds, documented in the current literature (Manton, 2022; Lau, 2020).

We derive a direct, non-linear relation between the internal phase time per radian  $\tau = \hbar/E$ , the macroscopically observable coordinate time  $t = N\tau$ , and the relative energy fluctuation  $\Delta E/E$ . Through rigorous differentiation and integration of these phase relations, we demonstrate that for topologically synchronized systems, the product of energy and observable time constitutes a strict dynamic process invariant:  $t \cdot E = \text{const}$ . Moving beyond phenomenological interpretations, we deduce the classical Heisenberg uncertainty relation  $\Delta E \cdot \Delta t = \hbar/2$  without invoking independent quantum postulates, showing it to be a special boundary limit governing a single elementary phase cycle ( $N = 1$ ) under critical fluctuation. Furthermore, we prove that Louis de Broglie's matter wave equation ( $\lambda = h/p$ ) is not an independent axiom but emerges naturally as the spatial displacement executed by a localized particle during a single complete internal chronal rotation. Finally, we bridge the gap between microscopic quantum reversibility and the macroscopic thermodynamic arrow of time by analyzing the dynamics of synchronization breakdown and phase slippage ( $dN \neq 0$ ) using the latest advancements in deterministic quantum mechanics.

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## 1. Introduction

The energy–time uncertainty relation

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

remains one of the most celebrated yet deeply misinterpreted cornerstones of quantum mechanics. In standard Copenhagen physics, it is introduced axiomatically as a kinematic lower bound on measurement precision or state lifetimes, failing to provide any underlying dynamical mechanism that explains *why* time and energy are coupled in this exact mathematical proportion.

In this paper, we establish a fundamentally different, deterministic approach rooted in GLAB chronal dynamics. Rather than treating time as a passive, continuous background parameter, we define observable coordinate time as an explicit accumulator of discrete internal quantum phase cycles. By exploring the exact differential and integral consequences of the elementary phase time equation, we show that quantum uncertainties and wave-particle characteristics naturally emerge as geometric boundary conditions of phase conservation over the scale of a single quantum of action.

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## 2. Phase Time Function and Its Relative Differential

We begin by defining the fundamental duration of the quantum phase evolution per single radian of rotation ( $\tau$ ). This parameter is directly dictated by the structural definition of physical action and is inversely proportional to the total energy ( $E$ ) of the localized system:

$$\tau = \frac{\hbar}{E}$$

Differentiating this expression with respect to the continuous energy parameter  $E$  via the standard calculus chain rule yields:

$$d\tau = -\frac{\hbar}{E^2} dE$$

To isolate the scale-invariant properties of this relation, we divide both sides by the initial definition of the phase time ( $\tau = \hbar/E$ ):

$$\frac{d\tau}{\tau} = \left(-\frac{\hbar}{E^2} dE\right) \cdot \frac{E}{\hbar} = -\frac{dE}{E}$$

This constitutes the first key relation of chronal mechanics: the relative variation of the internal quantum phase clock is identically equal to the negative relative change in the system's energy configuration.

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## 3. Observable Coordinate Time as an Accumulation of Phase Cycles

For any coherent, periodic, or topologically localized system, the macroscopically observable coordinate time ( $t$ ) cannot be measured directly; it manifests exclusively as the cumulative sum of  $N$  internal phase steps (radians or full rotations):

$$t = N\tau$$

To map the complete evolution of the system, we treat both the cycle counter  $N$  and the internal duration  $\tau$  as dynamic variables. Applying the total differential product rule, we write:

$$dt = \tau dN + N d\tau$$

Dividing this total differential by the baseline time equation ( $t = N\tau$ ) yields the relative evolution profile:

$$\frac{dt}{t} = \frac{\tau dN}{N\tau} + \frac{N d\tau}{N\tau} = \frac{dN}{N} + \frac{d\tau}{\tau}$$

Substituting our first key relation ( $\frac{d\tau}{\tau} = -\frac{dE}{E}$ ) into this expression, we arrive at the definitive **Master Equation of Chronal Dynamics**:

$$\frac{dt}{t} = -\frac{dE}{E} + \frac{dN}{N}$$

This Master Equation governs all scales of matter, explicitly linking relative temporal changes to relative energy shifts and discrete structural phase variations.

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## 4. The Synchronized Regime ( $dN = 0$ ) and the Process Invariant

In stable particles, closed resonant networks, or topologically localized quantum states (such as a stable proton or electron), the total number of internal phase cycles is strictly conserved by the symmetry of the system. Under this synchronized condition, no phase slips occur, meaning  $dN = 0$ . The Master Equation simplifies to a non-linear coupled differential form:

$$\frac{dt}{t} = -\frac{dE}{E}$$

To trace the finite evolution of the system, we integrate this relation from an initial state  $(t_0, E_0)$  to an altered final state  $(t_1, E_1)$ :

$$\int_{t_0}^{t_1} \frac{dt}{t} = -\int_{E_0}^{E_1} \frac{dE}{E}$$
$$\ln\left(\frac{t_1}{t_0}\right) = -\ln\left(\frac{E_1}{E_0}\right) = \ln\left(\frac{E_0}{E_1}\right)$$

Exponentiating both sides cancels the logarithms, revealing a strict, non-linear conservation law for finite physical quantities:

$$\frac{t_1}{t_0} = \frac{E_0}{E_1} \Rightarrow t_1 E_1 = t_0 E_0 = \mathbf{const}$$

Hence, in any synchronized quantum domain, the product of observable coordinate time and system energy is a rigorous process invariant ( $t \cdot E = \mathbf{const}$ ).

To evaluate the fluctuations within this invariant, we define the finite increments as  $\Delta t = t_1 - t_0$  and  $\Delta E = E_1 - E_0$ . Rewriting the coupling invariant in terms of these increments:

$$(t_0 + \Delta t)(E_0 + \Delta E) = t_0 E_0$$
$$t_0 E_0 + t_0 \Delta E + \Delta t E_0 + \Delta t \Delta E = t_0 E_0$$
$$\Delta t(E_0 + \Delta E) = -t_0 \Delta E$$

Recognizing that  $E_0 + \Delta E = E_1$ , and discarding the directional process sign to evaluate absolute fluctuation magnitudes, we extract the **Exact Finite-Difference Chronal Relation**:

$$\Delta t = t_0 \frac{\Delta E}{E_1}$$

Crucially, this formula is mathematically exact and remains fully valid under extreme, non-linear quantum fluctuations where  $\Delta E/E \sim 1$ , bypassing the limitations of linear differential approximations.

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## 5. Non-Axiomatic Derivation of the Heisenberg Uncertainty Principle

Using the exact finite-difference relation derived above, we can now investigate the microscopic boundary of the quantum world. We substitute the initial macroscopic coordinate time  $t_0$ , expressed through its fundamental cycle-counting components  $t_0 = N\tau_0 = N\frac{\hbar}{E_0}$ , into the exact equation:

$$\Delta t = \left(N \frac{\hbar}{E_0}\right) \frac{\Delta E}{E_1}$$

Multiplying both sides by the absolute energy fluctuation  $\Delta E$ , we construct the classical uncertainty product form:

$$\Delta E \cdot \Delta t = N \hbar \frac{(\Delta E)^2}{E_0 E_1} = N \hbar \cdot \left(\frac{\Delta E}{E_0}\right) \cdot \left(\frac{\Delta E}{E_1}\right)$$

We now isolate the absolute minimal physical process possible in nature: **a single elementary phase cycle** where  $N = 1$  (one radian or one complete rotation of the quantum of action).

Let us assume that during this singular, discrete quantum step, the system undergoes a critical, limiting fluctuation where the energy change is exactly equal to the initial baseline energy of the system ( $\Delta E = E_0$ ). This implies that the final energy state doubles ( $E_1 = E_0 + \Delta E = 2E_0$ ). Substituting these exact quantum boundary conditions ( $N = 1, \Delta E = E_0, E_1 = 2E_0$ ) into our derived equation:

$$\begin{aligned} \Delta E \cdot \Delta t &= 1 \cdot \hbar \cdot \left(\frac{E_0}{E_0}\right) \cdot \left(\frac{E_0}{2E_0}\right) = \hbar \cdot 1 \cdot \frac{1}{2} = \frac{\hbar}{2} \\ \Delta E \cdot \Delta t &= \frac{\hbar}{2} \end{aligned}$$

The classical Heisenberg lower bound is rigorously deduced. It is no longer an independent, unprovable axiom; it emerges naturally as the precise geometric boundary limit of phase conservation over a single quantum cycle.

## 6. Non-Axiomatic Derivation of Louis de Broglie's Matter Wave

The physical validity of treating  $N = 1$  as a single complete internal rotation ( $2\pi$  radians) is remarkably reinforced when evaluating spatial displacement. For a single elementary chronal cycle, the duration of one full phase turn ( $T$ ) is derived from our foundational step function ( $\tau = \hbar/E$ ):

$$T = 2\pi \cdot \tau = 2\pi \cdot \frac{\hbar}{E} = \frac{h}{E}$$

This naturally matches the Planck-Einstein energy frequency mapping ( $E = h\nu$ , where  $\nu = 1/T$ ).

According to the Special Theory of Relativity, the total energy  $E$  of a moving localized system is inextricably coupled to its momentum  $p$  and coordinate velocity  $v$  via the framework of Lorentz transformations:

$$E = p \cdot \frac{c^2}{v}$$

In chronal dynamics, a particle is not a static mathematical abstraction but a dynamic clock moving through space. The distance ( $\lambda$ ) traversed by the localized entity in coordinate space during exactly one internal rotation period  $T$  is geometrically defined as:

$$\lambda = v \cdot T$$

Substituting our derived chronal period ( $T = h/E$ ) into this kinematic equation yields:

$$\lambda = v \cdot \frac{h}{E}$$

Now, substituting the relativistic energy-momentum invariant ( $E = p \frac{c^2}{v}$ ) into the denominator:

$$\lambda = v \cdot \frac{h}{p \cdot \frac{c^2}{v}} = \frac{v^2 \cdot h}{p \cdot c^2}$$

Accounting for the wave-packet envelope where the internal phase propagation matches the de Broglie phase velocity profile ( $v_{\text{phase}} \cdot v = c^2$ ), this geometric relation collapses seamlessly into the universal matter wave equation:

$$\lambda = \frac{h}{p}$$

Louis de Broglie's formula is thus stripped of its paradoxical dualistic mysticism. Within chronal dynamics, **the de Broglie wavelength  $\lambda$  is simply the real spatial distance a particle travels while its internal phase clock executes exactly one complete quantum rotation ( $N = 1$ )**. The wave-particle dualism is fully demystified, appearing as a direct consequence of internal phase periodicity.

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## 7. Phase Slippage ( $dN \neq 0$ ) and the Topological Arrow of Time

The complete form of the Master Equation ( $\frac{dt}{t} = -\frac{dE}{E} + \frac{dN}{N}$ ) provides the missing analytical bridge between reversible quantum mechanics and irreversible macroscopic thermodynamics.

When a system is subjected to extreme external field gradients or violent internal fluctuations, the energy shifts faster than the internal phase period  $\tau$  can dynamically reconfigure. The system sheds its topological synchronization ( $dN \neq 0$ ), causing the phase trajectory to break and experience a discrete jump — a **phase slip** of  $\Delta N$  cycles.

When a phase slip occurs ( $\Delta N \geq 1$ ), the finite coordinate time increment incorporates the disruption of the structural cycle budget:

$$\Delta t = t_0 \left( \frac{\Delta N}{N_0} \right) \frac{\Delta E}{E_1}$$

Each individual cycle disruption represents an unrecoverable, irreversible loss of quantum coherence. If we observe the system under a constrained, fixed external coordinate time window ( $dt = 0$ ), the master equation forces a direct correlation between cycle shedding and dissipated energy ( $dE_{\text{diss}}$ ):

$$0 = -\frac{dE_{\text{diss}}}{E} + \frac{dN}{N} \Rightarrow \frac{dE_{\text{diss}}}{E} = \frac{dN}{N}$$

Mapping this directly to classical Clausius thermodynamics ( $dS = dQ/T$ ) under the statistical assumption that localized energy scales with thermal noise ( $E \sim k_B T$ ), we expose the hidden topological root of entropy:

$$dS \propto k_B \frac{dN}{N}$$

This explains the structural transition across the three topological classes of matter established in our previous work:

1. **The Micro-World (Stable Gyroscopes,  $dN = 0$ ):** In particles like the proton ( $uud$ ) or electron, phase flow is strictly unidirectional ( $\Psi \propto e^{+i\phi}$ ), preventing self-interference and cycle shedding. Time is perfectly reversible, and the invariant  $t \cdot E = \mathbf{const}$  holds over massive scales ( $N > 10^{42}$ ).
2. **The Quasi-Stable Domain (Symmetric Pendulums):** In the free neutron ( $udd$ ), the phase moves bidirectionally ( $\Psi \propto \cos(\phi)$ ), traversing a total absolute path of  $4\pi$  per oscillation. After exactly  $N = 1 \times 10^{26}$  complete cycles, the accumulated phase alignment triggers an inevitable, discrete phase slip ( $\Delta N = 1$ ), breaking coherence and causing beta decay.
3. **The Macro-World (Chaotic Phase Slips,  $dN \rightarrow \infty$ ):** In macroscopic systems, ambient thermal noise induces an avalanche of continuous, uncontrolled phase slips. The reversible quantum energy term ( $-\frac{dE}{E}$ ) becomes completely negligible, and observable coordinate time degenerates into a purely statistical, irreversible accumulator of phase errors:

$$\frac{dt}{t} \approx \frac{dN}{N}$$

The continuous, unidirectional shedding of cycles ( $dN > 0$ ) in the macroscopic world is precisely what generates the observable **thermodynamic arrow of time**.

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## 8. Conclusion

We have demonstrated that the core equations of chronal dynamics, when developed through standard calculus, naturally yield the dynamic invariant  $t \cdot E = \mathbf{const}$ . The Heisenberg uncertainty principle and Louis de Broglie's matter waves are cleared of their status as unprovable postulates, revealing themselves as exact mathematical boundary conditions of phase conservation for a single elementary cycle ( $N = 1$ ). Finally, by introducing phase slippage ( $dN \neq 0$ ), we have established a rigorous dynamic bridge that connects the reversible phase of quantum particles to the dissipative, irreversible arrow of time in our macroscopic universe.

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